HIRING PROCESS ANALYTICS

PROJECT DESCRIPTION:

As we all are very much aware of hiring process in today's world and how crucial role it plays in one's life, it could be any kind of hiring in any field and every other field has their own pattern of hiring it differs from one field to others as they are millions of different roles in different fields

As of now, me being Data analyst at a multinational company like google, my task would be to analyse the company's hiring process data and filter out all the important insights from it so that it could be very easier task for the hiring department. whether to know number of vacancies left which need to be filled with some certainties or urgent hiring or salary analysis or distribution, departmental analysis or position tier analysis etc…

A well-designed hiring process ensures that the best candidates are selected for the job, ensuring growth & welfare of organisation because it directly impacts the success, culture & overall performance of an organisation

APPROACH

My approach towards this project would be very simple and easily understandable, As we've been provided with a relevant dataset which contains all the data regarding hiring process including application id, interview time, gender specification, status, event name, different departments & posts, salary distribution

Now our work could include data analysing, data cleaning & data visualizing representing data differently as bar graphs, chats using pivot tables and getting the required information using various formulas

TECH STACK USED

I've used Microsoft excel 2022 for this project, It offers a wide range of data analysis functionalities, efficient data management, data visualization (as in charts, graphs etc..), powerful data manipulation tools etc.…

As Microsoft excel is very essential for our day-to-day life especially for people who work in any working field it has been widely used by everyone from schools to colleges to companies to personal or professional work purposes, whether you're an individual managing personal finances, a small business owner tracking sale, or a data scientist analysing large datasets, Excel is a crucial tool for organizing, analysing, and visualizing data. Its combination of simplicity and complexity makes it both accessible for beginners and powerful enough for experts.

INSIGHTS

One of the reasons about why giving insights are important cause it could potentially help the company improve its hiring process and make better hiring decisions in the future.

Some of my useful insights includes:

- Firstly, I've stared with identifying blank cells, missing values & duplicate values in the given data set in each and every column
- Filtered, sorted & replaced with required data if needed
- Data cleaning and data visualization tools are major aspect of this project
- As per the tasks, first task was about, I need to know like how many males and females got hired with using formulas
- Next task was to Simply salary analysis of the company by average using some excel functions
- Next, Salary distribution between class intervals could provide us with a frequency so that we could have an idea about how many are getting paid between ranges
- Then comes the Departmental analysis would be very beneficial as it is easy to get information regarding particular information through departments
- And finally Different positions within a company often have different tiers or levels.so analysing positional tiers was important to segregating based on their positions and also easily identifying

Here what I did:

Handling Missing Data: Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.

So firstly after downloading the dataset after going through the dataset, The dataset wasn't cleaned, It has some missing values, It has some blank spaces, has some outliers and also duplicate values so to analysis data we could face some problems which might effect our outcome and also it would be hard to understand as always to explain so in order to understand and analyse dataset to get the favourable outcomes we need to do some cleaning, altering, replacing ,changing etc…

Firstly to check whether I've any blank cells, there are many different ways to find out blank cells I've chosen one of known process according to my convenience, I've selected entire data with CTRL+A and then went to find & select tab clicked on go to specials then clicked on blank tab then I got to know any cell is blank

78	22155	07-05-2014 16:26	Rejected	Female	Finance Department	b9	46852
79	572422	07-05-2014 08:07	Rejected	Male	Sales Department	i7	61488
80	114584	07-05-2014 08:08	Rejected	Male	Sales Department	i7	
81	794226	07-05-2014 16:01	Rejected	Male	Service Department	i7	16090
82	496573	07-05-2014 16:01	Rejected	Male	Service Department	i7	83364
83	437695	13-05-2014 15:22	Rejected	Male	Service Department	i7	77517
84	983577	08-05-2014 07:13	Rejected	Male	Service Department	b9	84746
85	720850	08-05-2014 22:10	Rejected	Male	Sales Department	i6	80600

For handling missing data, there was some missing data with hyphen in post name column to know how many cells are with hyphen I've selected that column and select sort and filter tab then filtered that column by selecting any cell with missing value by that I got

	A	B	C	D	E	F	G	H
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary	
7	289907	01-05-2014 07:44	Hired	Male	Sales Department	-	85914	

And in the same way event_name column we have done the same process to know missing values and we got

	A	B	C	D	E	F	G	H	I	J
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary			
17	195323	09-05-2014 12:48	Hired	-	Service Department	i7	81757			
19	742283	02-05-2014 08:11	Rejected	-	Service Department	i5	100			
1602	227046	27-08-2014 18:08	Hired	-	Operations Department	b9	76730			
1791	711350	16-07-2014 13:33	Rejected	-	Operations Department	c-10	25785			
2878	835053	16-05-2014 18:34	Hired	-	Operations Department	c5	25583			
3259	444043	11-07-2014 14:52	Hired	-	Sales Department	c5	80262			
4018	352309	20-08-2014 10:38	Hired	-	Service Department	i5	4308			
4126	204014	09-08-2014 16:09	Rejected	-	Purchase Department	c5	96396			
4410	901867	18-08-2014 09:36	Rejected	-	Service Department	c5	22393			
5560	937905	08-08-2014 19:29	Hired	-	Marketing Department	c9	94032			
5607	564743	28-08-2014 10:25	Rejected	-	Production Department	c9	4076			
5889	245473	14-05-2014 18:48	Hired	-	Service Department	c5	66948			
6330	411295	22-06-2014 14:38	Hired	-	Operations Department	i1	98070			
6658	487617	30-05-2014 16:29	Hired	-	Service Department	c8	12470			
6998	827628	30-08-2014 15:51	Hired	-	Service Department	i1	3134			

So, these were all missing data and blank cells

Clubbing Columns: If there are columns with multiple categories that can be combined, do so to simplify your analysis

As per the given dataset they were no categories that could combine and could simplify our analysis but In the given dataset we got combined date and time in one column so we don't need any other alterations

Outlier Detection: Check for outliers in the dataset that may skew your analysis.

There are many ways for outlier detection, it could be sorting, visualizing, Interquartile range (IQR), Statistical procedures etc.......
So, there was a missing value in post name but we have hired him in sales department and offered salary of 85914 so here we can handle this outlier
As he's from sales department we need to filter sales department from department column and also need to filter his salary range too which is somewhere near to his range and by selecting that we can see use **Mode** and sort it with post name which offers similar salary range and we got

	A	B	C	D	E	F	G
1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salar
7	289907	01-05-2014 07:44	Hired	Male	Sales Department	-	85914
500	951001	22-06-2014 09:07	Hired	Female	Service Department	c-10	85827
814	978831	21-07-2014 17:02	Rejected	Male	Operations Department	i7	85868
1323	882740	07-08-2014 13:42	Rejected	Male	Sales Department	c9	85837
2043	597119	09-07-2014 10:10	Hired	Female	Service Department	i5	85877
2256	101190	09-05-2014 17:23	Hired	Male	Sales Department	c9	85057
3405	844634	22-07-2014 13:10	Hired	Male	Operations Department	c9	85902
4966	944862	24-06-2014 11:12	Hired	Female	Operations Department	i7	85882
6178	590453	14-06-2014 14:05	Hired	Male	Marketing Department	c5	85888

It's completely on us to change it to c9 because that might be the post name with that salary range

And in the post name column among all the names one looks kind of different i.e.(c-10) where others are without hyphen like c9, i5, c5 etc…
So to change it to with all others like c10 we can filter that column to c-10 then select that column click on find and select tab then find and their we need to find c-10 then find all andthen replace it with c10 and that's done

BEFORE ALTERING IT LOOKED LIKE THIS

1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary
142	361096	15-05-2014 09:56	Rejected	Male	Service Department	c-10	9390
143	691216	15-05-2014 09:56	Rejected	Male	Service Department	c-10	67066
144	567661	15-05-2014 09:57	Rejected	Male	Service Department	c-10	8723
145	382645	15-05-2014 09:57	Hired	Male	Service Department	c-10	65587
146	767003	15-05-2014 10:01	Hired	Male	Service Department	c-10	73396
147	412827	15-05-2014 15:57	Rejected	Male	Service Department	c-10	76789
151	105729	15-05-2014 16:13	Rejected	Male	Service Department	c-10	80817
173	303466	19-05-2014 13:17	Rejected	Male	Production Department	c-10	81257
174	549934	19-05-2014 13:18	Rejected	Male	Production Department	c-10	59735
178	299540	20-05-2014 17:03	Rejected	Female	General Management	c-10	98404
179	66952	20-05-2014 17:04	Rejected	Male	General Management	c-10	58443
180	372707	20-05-2014 17:05	Rejected	Female	General Management	c-10	92123
181	703540	20-05-2014 17:06	Hired	Female	General Management	c-10	77027
182	838911	20-05-2014 17:03	Hired	Male	General Management	c-10	98822
183	51314	20-05-2014 17:05	Hired	Female	General Management	c-10	18661
184	716333	20-05-2014 17:03	Hired	Don't want to say	General Management	c-10	71461
199	826902	21-05-2014 12:35	Rejected	Female	Finance Department	c-10	23823
209	150277	22-05-2014 22:18	Rejected	Male	Sales Department	c-10	91928
226	725295	27-05-2014 16:26	Rejected	Male	Operations Department	c-10	45985
228	328190	26-05-2014 14:51	Rejected	Male	Purchase Department	c-10	58037
240	417540	27-05-2014 19:43	Rejected	Male	Marketing Department	c-10	2121
241	180642	27-05-2014 19:44	Rejected	Male	Marketing Department	c-10	69642
259	13742	28-05-2014 01:42	Hired	Male	Production Department	c-10	40601
269	865692	28-05-2014 10:40	Rejected	Male	Operations Department	c-10	77581
274	67222	29-05-2014 18:46	Hired	Male	Operations Department	c-10	72545

AFTER ALTERING REPLACING C-10 WITH C10

1	application_id	Interview Taken on	Status	event_name	Department	Post Name	Offered Salary
142	361096	15-05-2014 09:56	Rejected	Male	Service Department	c10	9390
143	691216	15-05-2014 09:56	Rejected	Male	Service Department	c10	67066
144	567661	15-05-2014 09:57	Rejected	Male	Service Department	c10	8723
145	382645	15-05-2014 09:57	Hired	Male	Service Department	c10	65587
146	767003	15-05-2014 10:01	Hired	Male	Service Department	c10	73396
147	412827	15-05-2014 15:57	Rejected	Male	Service Department	c10	76789
151	105729	15-05-2014 16:13	Rejected	Male	Service Department	c10	80817
173	303466	19-05-2014 13:17	Rejected	Male	Production Department	c10	81257
174	549934	19-05-2014 13:18	Rejected	Male	Production Department	c10	59735
178	299540	20-05-2014 17:03	Rejected	Female	General Management	c10	98404
179	66952	20-05-2014 17:04	Rejected	Male	General Management	c10	58443
180	372707	20-05-2014 17:05	Rejected	Female	General Management	c10	92123
181	703540	20-05-2014 17:06	Hired	Female	General Management	c10	77027
182	838911	20-05-2014 17:03	Hired	Male	General Management	c10	98822
183	51314	20-05-2014 17:05	Hired	Female	General Management	c10	18661
184	716333	20-05-2014 17:03	Hired	Don't want to say	General Management	c10	71461
199	826902	21-05-2014 12:35	Rejected	Female	Finance Department	c10	23823
209	150277	22-05-2014 22:18	Rejected	Male	Sales Department	c10	91928
226	725295	27-05-2014 16:26	Rejected	Male	Operations Department	c10	45985
228	328190	26-05-2014 14:51	Rejected	Male	Purchase Department	c10	58037
240	417540	27-05-2014 19:43	Rejected	Male	Marketing Department	c10	2121
241	180642	27-05-2014 19:44	Rejected	Male	Marketing Department	c10	69642
259	13742	28-05-2014 01:42	Hired	Male	Production Department	c10	40601
269	865692	28-05-2014 10:40	Rejected	Male	Operations Department	c10	77581
274	67222	29-05-2014 18:46	Hired	Male	Operations Department	c10	72545

And also, there are some hyphens/ missing values in event_name too, to fix that I've filtered out hyphens, selected find & select, found all hyphens and replaced with" Don't want to say" as we already have that option in that because hyphen and don't want to say isn't different, has similar meaning

Before replacement:

application_id	Interview Taken on	Status	event_name
383422	01-05-2014 11:40	Hired	Male
907518	06-05-2014 08:08	Hired	Female
176719	06-05-2014 08:08	Rejected	Male
429799	02-05-2014 16:28	Rejected	Female
253651	02-05-2014 16:32	Hired	Male
289907	01-05-2014 07:44	Hired	Male
959124	06-05-2014 16:27	Rejected	Male
86642	09-05-2014 13:17	Rejected	Male
751029	02-05-2014 13:09	Hired	Female
434547	02-05-2014 13:11	Rejected	Female
518854	01-05-2014 09:00	Rejected	Male
649039	07-05-2014 10:48	Hired	Female
199526	07-05-2014 10:50	Hired	Male
539803	15-05-2014 09:31	Hired	Male
191009	09-05-2014 12:48	Hired	Female
195323	09-05-2014 12:48	Hired	-
51318	02-05-2014 08:07	Hired	Male
742283	02-05-2014 08:11	Rejected	-
513166	01-05-2014 22:53	Hired	Female
791372	01-05-2014 22:54	Rejected	Male
47857	01-05-2014 22:55	Rejected	Female
834101	01-05-2014 22:53	Rejected	Don't want to say
985008	01-05-2014 09:41	Rejected	Male
891568	01-05-2014 16:28	Hired	Female
935899	10-05-2014 14:17	Rejected	Male
780839	10-05-2014 14:18	Hired	Female

And after replacement:

application_id	Interview Taken on	Status	event_name
383422	01-05-2014 11:40	Hired	Male
907518	06-05-2014 08:08	Hired	Female
176719	06-05-2014 08:08	Rejected	Male
429799	02-05-2014 16:28	Rejected	Female
253651	02-05-2014 16:32	Hired	Male
289907	01-05-2014 07:44	Hired	Male
959124	06-05-2014 16:27	Rejected	Male
86642	09-05-2014 13:17	Rejected	Male
751029	02-05-2014 13:09	Hired	Female
434547	02-05-2014 13:11	Rejected	Female
518854	01-05-2014 09:00	Rejected	Male
649039	07-05-2014 10:48	Hired	Female
199526	07-05-2014 10:50	Hired	Male
539803	15-05-2014 09:31	Hired	Male
191009	09-05-2014 12:48	Hired	Female
195323	09-05-2014 12:48	Hired	Don't want to say
51318	02-05-2014 08:07	Hired	Male
742283	02-05-2014 08:11	Rejected	Don't want to say
513166	01-05-2014 22:53	Hired	Female
791372	01-05-2014 22:54	Rejected	Male
47857	01-05-2014 22:55	Rejected	Female
834101	01-05-2014 22:53	Rejected	Don't want to say
985008	01-05-2014 09:41	Rejected	Male
891568	01-05-2014 16:28	Hired	Female
935899	10-05-2014 14:17	Rejected	Male
780839	10-05-2014 14:18	Hired	Female

REMOVING DUPLICATE VALUES:

And now I've removed duplicate vales in application id column because all other columns definitely have repeated cells and those are mandatory too but in application id everyone has Id's which differ so I've selected application id column and selected conditional formatting and then clicked on duplicate values and it highlights the number with red colour as shown These are all duplicate values found

[illegible]	[illegible]
565233	2
596097	2
323756	2
788230	2
646167	2
155836	0
441985	2
44896	2
620118	2
249552	2
83479	2
859749	2
842011	3
264784	3
854925	3
607172	2
460382	2
986281	2
293153	2
985306	3
223490	3
869833	0
111712	3
503568	3

REMOVING OUTLIERS: Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.

Here I've selected salary offered column to detect outliers I've used box & whisker tab in charts cause other all looked confusing, the chart has represented below

We have to select data labels so that we could know where the salary offered lies
From this we get to know that
Higher bound is 99967
Lower bound is 100
Median is 49628
So here outliers are clearly shown which are 200000,300000& 400000 so I filtered out the offered salary column selected 200000,300000,400000&99967 (highest bound)
And I've replaced the outliers with highest bound and that might look like this

	B	C	D	E	F	G	H	I	J
1	Interview Taken on	Status	event_name	Department	Post Name	Offered Salar	After outlier removal		
13	07-05-2014 10:48	Hired	Female	Service Department	b9	200000	99967		
286	15-06-2014 09:45	Hired	Female	General Management	i4	400000	99967		
955	01-08-2014 10:47	Hired	Male	Service Department	c8	99967	99967		
6825	21-07-2014 15:39	Hired	Male	General Management	i7	300000	99967		

DATA SUMMARY:

After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data

Here I've my own descriptive, about this whole process as given in the project description, it was very essential for me to actually practically learn excel with help of sessions provided, and as well as use all of the tools, mostly essential ones, so that it enhances my practice and could lead me to actually become a great learner

So now some of my insights from the data would be:

- How identifying, analysing and cleaning data is very important
- How to find out blank cells through various process
- How to find missing values
- How to detect outliers
- How to remove duplicate values using different methods
- How to remove outliers through different strategies
- Usage of various visualization tools
- Usage of various features of tabs

With the help of sessions and as well as constant practice it was very helpful for data analysis

Data Analytics Tasks:

A. Hiring Analysis:

My Task: Determine the gender distribution of hires. How many males and females have been hired by the company?

Ans: Here in the given dataset, we've to know how many males & females got hired

I've used simple formula (i.e.) which is countifs and D:D is column where gender specification and C:C is either Hired or Rejected

For males: =COUNTIFS(D:D,"Male",C:C,"Hired") =2563

For females: =COUNTIFS(D:D,"Female",C:C,"Hired")=1856

GENDERS	HIRED
MALES	2563
FEMALES	1856

B. Salary Analysis:

My Task: What is the average salary offered by this company? Use Excel functions to calculate this.

Ans: Here we need to find out the average salary offered by the company, I've used sum of offered salary which is G:G column by count G:G then we got the average salary

Average salary= =SUM(G:G)/COUNT(G:G)= 49983.02902/49983.03

Average salary offered	49983.03

C. Salary Distribution:

My Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution**.**

Ans : Here I've created a table with two columns and first gave the class interval which is the difference between the upper and lower limits of a class and second column is frequency, which is salary range that falls between in that class interval using the COUNTIFS formula interval between upper and lower limit taken from salary range

=COUNTIFS(G:G,">=0",G:G,"<=10000")=678

=COUNTIFS(G:G,">=10001",G:G,"<=20000")=732

=COUNTIFS(G:G,">=20001",G:G,"<=30000")=711

=COUNTIFS(G:G,">=30001",G:G,"<=40000")=710

=COUNTIFS(G:G,">=40001",G:G,"<=50000")=782

=COUNTIFS(G:G,">=50001",G:G,"<=60000")=750

=COUNTIFS(G:G,">=60001",G:G,"<=70000")=698

=COUNTIFS(G:G,">=70001",G:G,"<=80000")=734

=COUNTIFS(G:G,">=80001",G:G,"<=90000")=711

=COUNTIFS(G:G,">=90001",G:G,"<=100000")=659

=COUNTIFS(G:G,">=100001",G:G,"<=200000")=1

=COUNTIFS(G:G,">=200001",G:G,"<=300000")=1

=COUNTIFS(G:G,">=300001",G:G,"<=400000")=1

Which made a grand total of **= 7168**

class interval	Frequency
0-10000	678
10001-20000	732
20001-30000	711
30001-40000	710
40001-50000	782
50001-60000	750
60001-70000	698
70001-80000	734
80001-90000	711
90001-100000	659
100001 -200000	1
200001-300000	1
300001-400000	1
GRAND TOTAL	**7168**

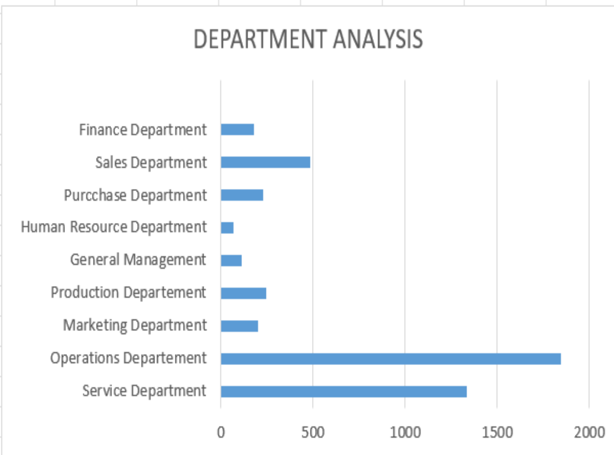
D. Departmental Analysis:

My Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

Ans : Here using visualization tools either charts or graphs or any other could use to represent proportion of people working in different departments
To make analysis I've used a formula for every department such as
Here we have all the department wise hired with the formula used countifs with department column and also with that hired/rejected column then we get result as follows

=COUNTIFS(E:E,"Service Department",C:C,"Hired")=1332
=COUNTIFS(E:E,"Operations Department",C:C,"Hired")=1843
=COUNTIFS(E:E,"Marketing Department",C:C,"Hired")=202
=COUNTIFS(E:E,"Production Department",C:C,"Hired")=246
=COUNTIFS(E:E,"General Management",C:C,"Hired")=113
=COUNTIFS(E:E,"Human Resource Department",C:C,"Hired")=70
=COUNTIFS(E:E,"Purchase Department",C:C,"Hired")=230
=COUNTIFS(E:E,"Sales Department",C:C,"Hired")=485
=COUNTIFS(E:E,"Finance Department",C:C,"Hired")=176

Department	Hired count
Service Department	1332
Operations Departement	1843
Marketing Department	202
Production Departement	246
General Management	113
Human Resource Department	70
Purcchase Department	230
Sales Department	485
Finance Department	176



E. Position Tier Analysis:

My Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Ans: Here we need to use any visualization tool for better understanding and also able to show the creative interesting way of showing data, we have filtered out all positions to analyse so we have used countif position column then we got the result as follows

=COUNTIF(F:F,"c8") =320
=COUNTIF(F:F,"c5")=1747
=COUNTIF(F:F,"b9")=463
=COUNTIF(F:F,"c10")=232
=COUNTIF(F:F,"c9")=1792
=COUNTIF(F:F,"i4")=88
=COUNTIF(F:F,"i5")=787
=COUNTIF(F:F,"i6")=527
=COUNTIF(F:F,"i7")=982
=COUNTIF(F:F,"m6")=3
=COUNTIF(F:F,"m7")=1
=COUNTIF(F:F,"n6")=1
=COUNTIF(F:F,"n9")=1
=COUNTIF(F:F,"n10")=1

Postion Tire	No of People
C8	320
C5	1747
B9	463
C10	232
C9	1792
I4	88
I5	787
I6	527
I7	982
M6	3
M7	1
N6	1
N9	1
N10	1